

Problem Definition:

Home field advantage has been talked about for decades with some people saying it is actually more useful than others. It has been talked about if you actually play better with home fans than being on the road with fans booing and talking bad to you. By looking at NFL game results from the last 10 years and calculating the home team's win percentage for each season, we can see whether it has changed over time or not. My research question I will be answering is "How Does Home-Field Advantage Vary Across NFL Seasons Throughout the Last 10 Years?" This question is relevant because it is an important part of understanding NFL games and team performance. NFL teams, players, coaches, fans, and analysts could all be interested in whether playing at home still provides a significant advantage.

Data Description:

The key variables I will be going over in this project include: Season, Home Team, Away Team, Home Score, Away Score, Game Result, Location, and Season Type. The way I am conceptualizing each one is first by looking at the years I will use for this data (2016-2025 (Season)), the team designated as the home/away team for a particular game (Home Team/Away Team), total number of points scored by the home/away team during a game (Home Score/Away Score), the outcome of the game from the perspective of the home team (Game Result), the physical venue or stadium where the game was played (Location), and the stage of the NFL season in which the game occurred (Season Type). The way I will operationalize them is by the name or standardized abbreviation of the team designated as the home/away team (Home Team/Away Team), total number of points scored at the end of the game by the home/away team (Home Score/Away Score), coded according to the outcome of the home team's perspective (Game Result), recorded as the stadium/venue that the game is played in (Location), and coded according to the stage of the season: Pre Season, Regular Season, Post Season. The data came from nflverse on GitHub, which is an open-source collection of NFL data and analytics resources. I used the games.csv dataset from the nflverse-data project. The nflverse project explains that its data is stored in GitHub releases, and the schedules release contains NFL game/schedule data. Each row represents one NFL game. The dataset contains games from every game from the years I filtered it by (2016-2025). I will group those games by season to determine how often the home teams win.

Links:

<https://github.com/nflverse/nflverse-data>

<https://github.com/nflverse/nflverse-data/releases>

<https://github.com/nflverse/nfldata/blob/master/data/games.csv>

Data Cleaning and Preparation:

For my data cleaning and preparation process, I first decided on what variables I was going to measure home field advantage on. I decided to do the 9 important variables which include: Season, Home Team, Away Team, Home Score, Away Score, Game Result, Location, and Season Type. I then focused on preparing my data for my project. I first printed a list of all of the original columns which was a total number of 46 columns:

```
print("Original columns:")
print(df.columns.tolist())
```

[19] ✓ 0.0s

...

ad_odds', 'total_line', 'under_odds', 'over_odds', 'div_game', 'roof', 'sun

The next thing I did was to tell the DataFrame to replace all of those columns with just the 8 variables that I am researching in this project:

```
df = df[
    [
        "season",
        "home_team",
        "away_team",
        "home_score",
        "away_score",
        "result",
        "location",
        "game_type"
    ]
]

print("\nColumns after cleaning:")
print(df.columns.tolist())
```

[20] ✓ 0.0s

...

Columns after cleaning:
['season', 'home_team', 'away_team', 'home_score', 'away_score', 'result', 'location', 'game_type']

After I was done cleaning all of the data that I was and wasn't going to use, I then went to set a name for all of the variables that I would be using in this research:

```
# Select the variables used in the project
final_df = pd.DataFrame({
    "Season": df["season"],
    "Home Team": df["home_team"],
    "Away Team": df["away_team"],
    "Home Score": df["home_score"],
    "Away Score": df["away_score"],
    "Location": df["stadium"],
    "Season Type": df["game_type"]
})

final_df.head()
```

[3] ✓ 0.0s

	Season	Home Team	Away Team	Home Score	Away Score	Location	Season Type
4515	2016	DEN	CAR	21.0	20.0	Sports Authority Field at Mile High	REG
4516	2016	ATL	TB	24.0	31.0	Georgia Dome	REG
4517	2016	BAL	BUF	13.0	7.0	M&T Bank Stadium	REG
4518	2016	HOU	CHI	23.0	14.0	NRG Stadium	REG
4519	2016	JAX	GB	23.0	27.0	EverBank Field	REG

The last preparation I had to do was to write a code to calculate the home field win percentage, and then filter it by season:

```
# Calculate home-team point differential
final_df["Point Differential"] = final_df["Home Score"] - final_df["Away Score"]

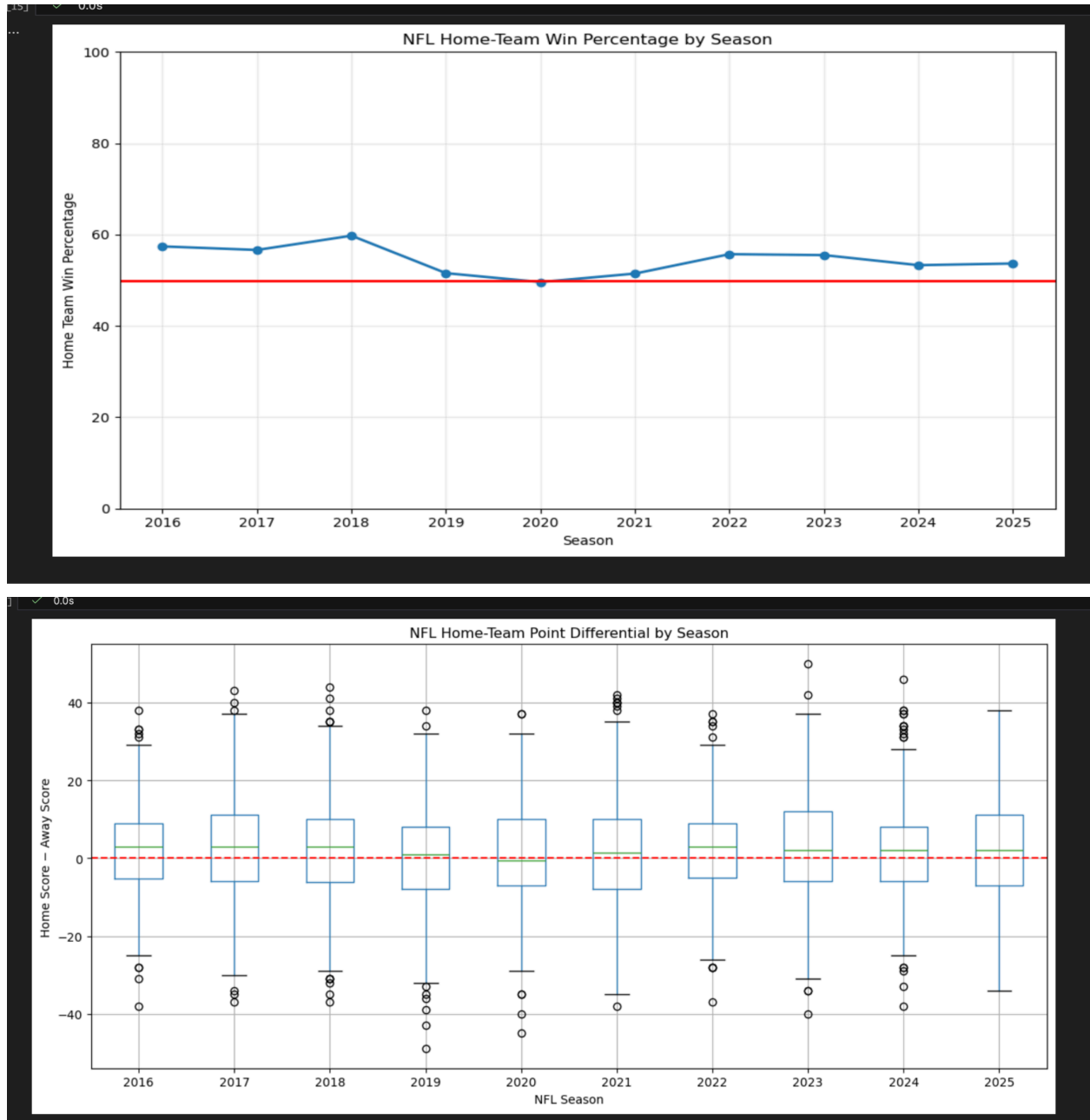
# Calculate home-team win percentage by season
home_win_percentage = (
    final_df.groupby("Season")["Game Result"]
    .apply(lambda x: (x == "Home Win").mean() * 100)
    .reset_index(name="Home Win Percentage")
    .sort_values("Season")
)

print(home_win_percentage.to_string(index=False))
```

[5] ✓ 0.0s

Season	Home Win Percentage
2016	57.421875
2017	56.640625
2018	59.765625
2019	51.562500
2020	49.609375
2021	51.470588
2022	55.719557
2023	55.514706
2024	53.308824
2025	53.676471

Data Understanding and Visualizations:



Storytelling and Narrative:

The visual analysis of the data portrays that there isn't a current trend with NFL teams playing at their home fields, but the average win percentage of home teams are above 50% besides 2020 where it was around 49.6%. It would be inaccurate to assume that there is no advantage to playing as the home team, as the results show win percentages above 50%.

Home field advantage is a big topic in the NFL talks that also have many factors going into the results such as their record going into the game, the attendance depending on weather, etc.

Limitations, Ethics, and Reflection:

The dataset does fail to include the record of the home/away teams going into each week, which I think would bring better results into multiple researches like the attendance percentage of NFL games, the win percentage of being at home, etc. Some biases that exist in the dataset could include when in the season the games happened like the post season, which has a completely different feel and atmosphere than playing a regular season game, which is why I filtered the games to be regular season so that it makes a more neutral feel. Also, the NFL has changed over the last 10 years with changes in rules, scheduling, traveling, and stadiums, meaning differences in seasons aren't just based on one factor. If I had more time or data to explore around this subject, I would want to factor in a couple more things such as what the weather forecast was that day of the game, the record of the home team going into the game because that plays a major factor in the attendance of those games.

Code Repository: